EngineAI T800 MuJoCo Simulation Diagnostic & Technical Guidance Report

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To: Guilherme Andrade
From: EngineAI Technical Support Team
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Subject: Technical Resolution for T800 MuJoCo Simulation Instability and Control Command Mapping

1. Root Cause Analysis

Based on your integration report, we have analyzed the issues regarding structural standing instability (early termination due to low CoM) and missing control commands during PPO policy training in MuJoCo:

Issue Phenomenon	Primary Root Cause	Underlying Mechanism
Missing Control Commands	Classical joint parser configuration mismatch	T800 employs proprietary drive structures in the legs and torso. Direct serial joint actions from RL bypass kinematic mapping if the parser flag is not configured, resulting in unmapped joint commands.
Stand-up & Stability Failure	Bypassing state machine initialization flow	Direct policy injection without initial ground-contact balance causes severe instability. Bypassing the Task Manager disables command smoothing and torque limiting, triggering CoM safety thresholds.

2. Integration Guidance & Resolution

2.1 Correct Integration via Native SDK (MotionRunner)

Do not bypass the Native SDK state machine. Implement your PPO policy as a custom MotionRunner so commands pass through Task Manager safety checks:

- **Create Custom Class:** Inherit from MotionRunner and register via

REGISTER_RUNNER(MyPPORunner, "ppo_runner", kMotion).

- **Configure Joint Parser Flag:** If your model outputs serial joint representations, disable default parallel parsing inside SetupContext():

```
data_store_>parallel_by_classic_parser.store(false);
```

- **Model Inference:** Load your exported .mnn model using math::MNNModel in Enter(), and extract observations during the 500 Hz control loop.
- **Command Output:** Set target joint actions via SetCommand() to apply PD gains, smoothing, and torque limits:

```
GetMutableOutput().SetCommand(q_des, qd_des, Kp, Kd, tau_ff_des);
```

2.2 Standardized Initialization Flow

Before executing your RL policy, perform the standard standing sequence:

1. Send LB+A (or virtual gamepad mapping Q+J) to initiate pd_stand mode.
2. Confirm ground contact and upright balance.
3. Transition to your custom ppo_runner policy.

2.3 Confirmation on Latest Repository Status

Please note that the current model files and SDK configurations available on our official GitHub repository (engineai-robotics/engineai_robotics_native_sdk) are fully up to date and contain the verified t800.xml kinematic definition and actuator parameters matching real hardware specs (Q300H, Q200H, Q50H, Q25H). No additional software patch or modified URDF/XML file is required.
Following the integration workflow detailed above—specifically configuring the joint parser flag and wrapping the policy inside a MotionRunner—will fully resolve the command mapping and stabilization issues in MuJoCo.